



Surveying endo-lysosomal function in PD subtypes using an LC-MS-based 12-plex in CSF

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Summary

PPMI CSF samples were analyzed for a series of lysosomal and lysosomal-related proteins. Specifically, the assay used here is a bottom up-liquid chromatography-mass spectrometry (LC/MS) which provides absolute quantitation of Cathepsin B, D, F, L, S, GCase, GM2A, GPNMB, LAMP1, LAMP2, LC3 and ubiquitin. Measurement of these analytes was performed across 3 time points (BL, year 1, year 2). The goal is to determine whether genotype and/or disease progression influences levels of any of the measured proteins.

Method

Sample Preparation and Digestion

A multiplexed LC-MS method for the detection of 12 lysosome-related proteins in human cerebrospinal fluid (CSF) was developed. Equal 25 μ L volumes of each CSF sample was spiked into equal 25 μ L volumes of artificial CSF (In house prepared @ Biogen, 125mM NaCl-26mM NaHCO₃, 1.25 mM NaH₂PO₄, 2.5mM KCL). Samples were then spiked with 2pg/mL of isotope labeled peptides TITLEVEPSDTIENVK_{13C(6)15N(2)} for Ubiquitin, LLMLDDQR_{13C(6)15N(2)} for Gcase, AYPVIAQVK_{13C(6)15N(2)} for GPNMB, IPLNDLFR_{13C(6)15N(2)} for LAMP2, AFSVNIFK_{13C(6)15N(2)} for LAMP1, EVAGLWIK_{13C(6)15N(2)} for GM2A, IPVIER_{13C(6)15N(2)} for LC3, VSTLPAITLK_{13C(6)15N(2)} for CatD, FSDLTEEEFR_{13C(6)15N(2)} for CatF, DIMAEIYK_{13C(6)15N(2)} for CatB, NSWGEEWGMGGYVK_{13C(6)15N(2)} for CatL and GIDSDASYPYK_{13C(6)15N(2)} for CatS as internal standards. Samples then spiked with 25 μ L 50:50 acetonitrile (OmniSolv[®] LC-MS grade, AX0156) and water (OmniSolv[®] LC-MS grade, WX0001), 8 μ L of 0.5mg/ml L-Glutathione (Sigma Aldrich, **G4251**) and 25 μ L of 150ug/ml bulk trypsin (Worthington bulk trypsin TPCK Treated CAT#LS003741) in a neat aCSF. Samples then digested for overnight at 37°C.

Solid Phase Extraction

Solid phase extraction was performed by using Oasis HLB 96-well μ Elution Plate (Waters 186001828BA). The plate was conditioned with 500 μ L of 0.1% formic acid in acetonitrile, followed by equilibration using 500 μ L of 0.1% formic acid in water. The 130 μ L sample was loaded onto the Oasis HLB 96-well μ Elution Plate, followed by a wash with 500 μ L of 5% methanol in water acidified with 0.1% formic acid. The peptides were eluted using 25 μ L of 35% acetonitrile in water acidified with 0.1% formic acid. The eluent sample was diluted to 250 μ L with 0.1%formic acid in water.





LC-MS Analysis of Lysosomal Panel Target Peptides

Diluted samples (20 μ L) were loaded onto an Evotip Pure and chromatographically separated using an Evosep One coupled to V1064 analytical column (100 μ m ID, 3 μ m, x 8 cm; Catalog # EV1064). Mass spectrometry was performed on a Thermo Exploris 240 operating in PRM mode. System was set to positive mode with a resolution of 12,500, AGC target at 5×10^5 , maximum injection time at 55 ms, and isolation window at 2 m/z. The table below specifies the m/z values and optimized HCD collision energy settings for both native and internal standards. Quantitation and data analysis were performed in SkyLine 24.1.0.199.

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